

Comparing Satyajit Ray Akira Kurosawa Directed Movies (#4850)

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1) What's the main question being asked or hypothesis being tested in this study?

Theoretical hypothesis

My theoretical hypothesis is that Satyajit Ray (from Kolkata, India) and Akira Kurosawa (from Japan) are both great directors, both won the academy award for lifetime achievement. Because they are both great, the movies they directed are equally good.

2) Describe the key dependent variable(s) specifying how they will be measured.

Dependent Variables to be measured

The dependent variables to be measured are the IMDB ratings (scores), # Users rated each movie.

First IMDB search will be used separately for the two legendary directors separately to get all the hits.

Then the search results will be sorted based on the release date, and 29 most recent full movies (excluding documentaries / TV series) will be used.

So in my case, I will use the 29 last movies Satyajit Ray and Akira Kurosawa directed in from today (excluding documentaries / TV series), the moment I did the IMDB search.

3) How many and which conditions will participants be assigned to?

Two conditions: Movies directed by Satyajit Ray & Akira Kurosawa

4) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

NHST and POST, 90% CI calculation on the effect size

5) Any secondary analyses?

No

6) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Justify the sample size

I predict no difference, and thus I'll do a power analysis for an equivalence test. I want to be pretty sure I can reject my smallest effect size of interest, so I'll design a study with 84% power. For this educational assignment, we do not collect a huge amount of data.

As long as I can exclude a large effect (Cohen's $d=0.8$ or larger) I will be happy for this assignment.

The power analysis estimates that the sample size we need to show the difference between the ratings for movies directed by Satyajit Ray and Akira Kurosawa is smaller than Cohen's $d=0.8$ (assuming the true effect size is 0, and with an α of 0.05, when I aim for 84% power) is 29 movie ratings from Satyajit Ray, and 29 movie ratings from Akira Kurosawa, as can be seen from the following R code and the figures.

The α -level I found acceptable is 0.05.

I performed a two-sided test.

I used 84% power for this study.

The effect size expected is $0.78948 < 0.8$, as shown below.

Given that Satyajit Ray has total 29 full movies directed, we can only collect 29 observations for both the directors.

7) Anything else you would like to pre-register? (e.g., data exclusions, variables collected for exploratory purposes, unusual analyses planned?)

No

8) Have any data been collected for this study already?

No, no data have been collected for this study yet

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